

Reinforcement learning for geometry

Lecture 4 | AI meets geometry

XXII Escola de Geometria Diferencial – Teresina, Piauí

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Two parts

Part I — RL for mathematics

- From *labels* (Lectures 2–3) to *rewards*.
- What RL is, and why it fits discovery.
- Guiding example: *geodesics by search*.
- *Hands-on*: find a geodesic on a surface.

Part II — Line arrangements

- Freeness, Saito, Terao's conjecture.
- Saito's criterion $\rightarrow$ a continuous reward $\mathfrak{S}$.
- Playing the game with PPO; verify over $\mathbb{Q}$.
- A research case study — *expository*.

Part I: ~ 20 min discussion + a ~ 10 min notebook. **Part II:** ~ 30 min, expository.

Guiding theme. *AI proposes candidates; rigorous methods dispose.*



The image shows the homepage of the XXII Escola de Geometria Diferencial (EGD) website. The header includes the logo 'XXII EGD' and the URL 'https://egd.ufpi.edu.br/'. The navigation menu at the top right contains links for 'HOME', 'ABOUT', 'SCIENTIFIC PROGRAM', 'BOOKS', and 'REGISTRATION'. A red arrow points to the 'BOOKS' link. The main content area features a large graphic of stacked plates and the text 'XXII Escola de Geometria Diferencial' with the dates 'Teresina, July 13-17, 2026'. Social media icons for Instagram and Facebook are present, along with a 'Register Now' button. The footer contains the text 'Geometria Diferencial | XXII Escola de Geometria Diferencial | XXII Escola'.

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Brief Description

"AI meets Geometry" is a hands-on minicourse introducing modern artificial intelligence as a practical tool for exploration, computation, and discovery in differential geometry. Participants will learn how neural networks, physics-informed methods, optimization, and reinforcement learning can be used to approximate geometric structures, study metrics and differential forms, investigate geometric flows, and search for new examples and conjectures, while keeping clear the essential distinction between numerical discovery and mathematical verification. No prior machine-learning background is assumed: the course builds conceptual foundations alongside concrete geometric case studies, including special metrics, neural PDE methods, and AI-assisted discovery. Code will be provided throughout, giving attendees working notebooks that they can modify, extend, and adapt directly to questions arising in their own research.

Download



Notebooks

[README](#) [MIT license](#)

Lecture	Notebook	Open in Colab
Lecture 2	Part I	Open in Colab
Lecture 2	Part II	Open in Colab
Lecture 3	Full notebook	Open in Colab
Lecture 4	Part I	Open in Colab

AI meets Geometry

XXII ESCOLA DE GEOMETRIA DIFERENCIAL
FOUR-LECTURE MINICOURSE

Learn how AI can approximate geometric structures, solve variational PDEs, and search for new mathematical examples.

1 AI Overview

Map supervised, unsupervised, and reinforcement learning onto geometric approximation, pattern-finding, and search.

2 Learning Geometric Structures

Train neural networks to learn metrics and differential forms while enforcing symmetry, positivity, and geometric identities.

3 Neural PDEs & Geometric Flows

Encode geometric PDEs in neural losses and flow an immersed torus toward the Willmore minimiser.

4 Discovery & Verification

Search line arrangements and other geometric spaces with reinforcement learning, then verify discoveries exactly.

Packages

No packages published
[Publish your first package](#)

Contributors 2

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Languages

Jupyter Notebook 100.0%

Part I — Reinforcement learning for mathematics

Computers in mathematics: three roles

Verify

Check a *given* proof.

- Lean, Coq, Isabelle.
- Four-Colour, Kepler's Conjecture, Liquid Tensor.
- Answer: “*valid.*” or “*invalid.*”

Solve

Produce a proof of a stated problem.

- AlphaGeometry / AlphaProof: IMO silver, 2024.
- FirstProof project.
- Benchmarks of unpublished research lemmas.

Discover (today)

Suggest new patterns, examples, conjectures.

- Davies et al., *Nature* 2021.
- AlphaTensor, *Nature* 2022.
- *No statement to prove.*

How can a computer help us *discover*, not merely verify?

From labels to rewards

(Self-)Supervised

Lectures 2–3

- Data $\{(x_i, y_i)\}$; learn $f : x \mapsto y$.
- Needs **labels**.
- Euler's formula; the CY metric.

Unsupervised

- Data $\{x_i\}$; find structure.
- No labels.
- Clustering, dim. reduction.

Reinforcement (today)

- No fixed data; an agent *acts*.
- A **reward** signal.
- Search, games, construction.

In mathematics we usually have a clear *target property* to satisfy — which is exactly a reward.

$\Rightarrow$ many mathematical **search** problems are RL problems.

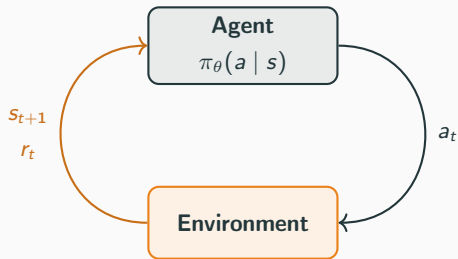
What is reinforcement learning?

An **agent** interacts with an **environment**:

- observe a state s_t ;
- pick an action $a_t \sim \pi_\theta(a \mid s_t)$;
- receive a reward r_t ;
- move to s_{t+1} .

Learn π_θ maximizing the expected return

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_t \gamma^t r_t \right].$$



A game with rules and a score. (an MDP)

Why RL fits mathematical discovery

Many searches share one shape:

- the object is built **step by step** (steps of a path, lines, generators, moves);
- there is a target property (a shortest path, freeness, a bound to beat);
- we can *measure progress* — a proxy — even without a proof.

actions + terminal condition + reward.

Our hands-on example: **geodesics by search** (Part I).

Then a research case study — **free line arrangements** (Part II, expository).

A guiding example: finding geodesics

Build a path, one step at a time

On (M, g) , fix endpoints p, q . Discretize a curve into nodes $p = x_0, x_1, \dots, x_N = q \in M$.

As a Markov decision process:

- **state** — the nodes placed so far;
- **action** — place the next node on M ;
- **reward** — minus the path's discrete energy $r = -\sum_i \ell(x_i, x_{i+1})^2$, on the *finished* path.

The return is $-E[\gamma] = -\frac{1}{2} \int_0^1 \|\gamma'\|_g^2 dt$: maximizing it *is* the classical variational problem, whose extremals are **geodesics** ($\nabla_{\gamma'} \gamma' = 0$). We reward energy, not length: length is reparametrization-invariant — energy also pins the *constant-speed* parametrization. RL as sampling-based calculus of variations.

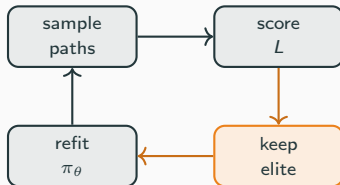
The simplest solver: the cross-entropy method

A small policy π_θ proposes paths (the node coordinates). Then loop:

1. **sample** a batch of paths from π_θ ;
2. **score** each by its length L ;
3. keep the **elite** — the shortest $\rho\%$;
4. **refit** π_θ to imitate the elite.

The policy concentrates on short paths — the geodesic.

The lightest reinforcement method — Part II's case study uses its policy-gradient cousin, **PPO**.



Rediscovery, then discovery

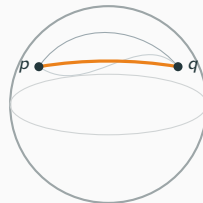
Sanity — a known answer.

On the round sphere S^2 the search recovers the great-circle arc between p, q ; its length $\rightarrow$ the central angle $\arccos\langle p, q \rangle$ — **exactly checkable**. (You'd just integrate the ODE — which is why it makes a perfect test.)

Discovery — no closed form.

On a triaxial ellipsoid geodesics have no elementary formula; the same loop finds them, and we *certify* by the residual $\nabla_{\gamma'} \gamma' \approx 0$.

At research scale: the same discipline — relax the geometric target, then optimize — learned the Calabi–Yau metric of Lectures 2–3.



S^2 : sampled paths (grey) collapse onto the great circle (orange).

Propose by search $\rightarrow$ certify by the geometry.

The template generalizes

AlphaTensor (2022)

- $\text{matmul} \leftrightarrow$ low-rank tensor;
- action: subtract a rank-1 term;
- found *faster exact* algorithms.

Learning to unknot (2021)

- state: a braid word;
- action: a Markov move;
- simplifies hard unknots.

Ribbon knots (2025)

- build a ribbon disk;
- new examples relevant to SPC4.

Same skeleton — state, action, reward. The game changes; the framework does not.

Discovery vs. verification

Discovery — cheap, fallible

Optimization / RL / heuristics navigate huge spaces and *propose* candidates, patterns, conjectures.

Verification — exact, final

Symbolic / exact computation and classical proof *certify* or refute.

Numerical or AI evidence is **not** a proof.

The discipline: *relax* the target into a searchable reward, then *certify* the winners exactly.

In the Part I practical, you will run this loop yourself.

Hands-on (Part I)

Find a geodesic by search: sample paths from a small policy, score by length, keep the shortest, refit — and watch them collapse onto the great-circle geodesic on S^2 , length $\rightarrow \arccos\langle p, q \rangle$.

(~10 min · notebook to follow)

Part II — Line arrangements: freeness as a game

Line arrangements in $\mathbb{P}^2$, and their combinatorics

Let $S = \mathbb{C}[x, y, z]$. A projective line arrangement is

$$\mathcal{A} = \{L_1, \dots, L_n\}, \quad L_i = V(\alpha_i), \quad \alpha_i \in S_1, \quad Q(\mathcal{A}) = \prod_{i=1}^n \alpha_i \in S_n.$$

Combinatorial data: the **intersection lattice** $\mathcal{L}(\mathcal{A})$. For a point p on m_p lines, $t_m = \#\{m\text{-fold points}\}$, and

$$\sum_{m \geq 2} \binom{m}{2} t_m = \binom{n}{2}, \quad b_2(\mathcal{A}) = \sum_p (m_p - 1) = \sum_{m \geq 2} (m - 1) t_m.$$

b_2 is a cheap, purely combinatorial invariant — later a constraint *and* a feature.

Let $\text{Der}_{\mathbb{C}}(S) = S\partial_x \oplus S\partial_y \oplus S\partial_z$. The module of *logarithmic derivations* of $\mathcal{A}$ is

$$D(\mathcal{A}) = \{\theta \in \text{Der}_{\mathbb{C}}(S) : \theta(\alpha_i) \in S\alpha_i \ \forall i\} = \{\theta : \theta(Q) \in (Q)\}.$$

Its elements are polynomial vector fields *tangent to every line* of $\mathcal{A}$.

- The *Euler derivation* $\theta_E = x\partial_x + y\partial_y + z\partial_z$ always lies in $D(\mathcal{A})$, since $\theta_E(Q) = nQ$.
- Via syzygies of the Jacobian ideal $J_Q = (Q_x, Q_y, Q_z)$:

$$D(\mathcal{A}) \cong S\theta_E \oplus \text{syz}(J_Q).$$

Definition

$\mathcal{A}$ is **free** if $D(\mathcal{A})$ is a free S -module. Then $D(\mathcal{A}) \cong S(-1) \oplus S(-d_1) \oplus S(-d_2)$, with exponents $(1, d_1, d_2)$.

Theorem (Saito, 1980)

$\mathcal{A}$ is free with basis $\{\theta_0, \theta_1, \theta_2\}$ iff $\det M(\theta_0, \theta_1, \theta_2) = c Q$ for some $c \in \mathbb{C}^\times$.

Taking $\theta_0 = \theta_E$ leaves the polynomial identity

$$\boxed{\det M(\theta_E, \theta_1, \theta_2) = c Q} \quad (\deg \theta_j = d_j).$$

Terao's factorization and candidate exponents

Terao (1981)

If $\mathcal{A}$ is free with exponents $(1, d_1, d_2)$, its characteristic polynomial factors:

$$\chi(\mathcal{A}, t) = (t - 1)(t - d_1)(t - d_2).$$

Comparing with $\chi(\mathcal{A}, t) = (t - 1)(t^2 - (n - 1)t + b_2 - n + 1)$,

$$d_1 + d_2 = n - 1, \quad d_1 d_2 = b_2(\mathcal{A}) - (n - 1).$$

A cheap necessary test.

$$\Delta(\mathcal{A}) = (n - 1)^2 - 4(b_2 - n + 1) \geq 0 \text{ and a perfect square.}$$

When it passes, (d_1, d_2) are the **candidate exponents** — necessary, far from sufficient.

Terao's conjecture

Terao's conjecture

Freeness of $\mathcal{A}$ is determined by its intersection lattice $\mathcal{L}(\mathcal{A})$.

Status.

- Open even for line arrangements in $\mathbb{P}^2$.
- Verified up to $n = 14$ (Barakat–Behrends–Jefferson–Kühne–Leuner, 2021) by exhaustive matroid enumeration; infeasible beyond.

The practical obstacle. Freeness is highly non-generic and configuration space grows fast — building explicit free arrangements is hard.

⇒ we want **automatic** ways to produce examples.

From a binary test to a continuous functional

Standard Saito. $\det M(\theta_E, \theta_1, \theta_2) \stackrel{?}{=} c Q$ — holds or it doesn't.

Set-up. A degree- d derivation $\theta = f\partial_x + g\partial_y + h\partial_z$ has $3N_d$ coefficients ($N_d = \binom{d+2}{2}$). Tangency to each line is $d + 1$ linear equations; stacking gives the *derivation matrix* $M_d \in \mathbb{R}^{n(d+1) \times 3N_d}$, and an orthonormal basis V_j of $\ker M_{d_j}$ parametrizes degree- d_j derivations $\theta_j = V_j \alpha_j$.

The determinant is bilinear. Expanding along the Euler row and substituting $\theta_j = V_j \alpha_j$ gives a tensor

$$T \in \mathbb{R}^{N \times k_1 \times k_2}, \quad T(\alpha_1, \alpha_2)_\beta = \sum_{i,j} T_{\beta ij} (\alpha_1)_i (\alpha_2)_j.$$

With q the coefficient vector of Q : $\text{Saito} \iff \exists \alpha_1, \alpha_2, c \neq 0 : T(\alpha_1, \alpha_2) = c q$.

The Saito functional

_____, *A semicontinuous relaxation of Saito's criterion and freeness as angular minimization*,
arXiv:2604.02995

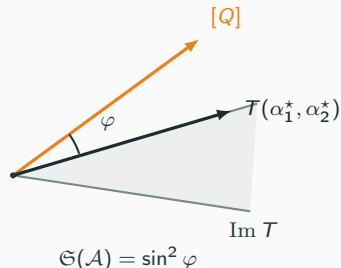
Definition

$$\begin{aligned}\mathfrak{S}(\mathcal{A}) &= 1 - \sup_{\alpha_1, \alpha_2 \neq 0} \frac{\langle T(\alpha_1, \alpha_2), q \rangle^2}{\|T(\alpha_1, \alpha_2)\|^2 \|q\|^2} \\ &= \inf \sin^2 \angle(T(\alpha_1, \alpha_2), q).\end{aligned}$$

The squared sine of the smallest angle between $\text{Im } T$ and the direction $[Q]$.

Theorem (_____, 2026)

- $\mathfrak{S}(\mathcal{A}) \in [0, 1]$; $\mathfrak{S}(\mathcal{A}) = 0 \iff \mathcal{A}$ free with exponents (d_1, d_2) ;
- upper semicontinuous on strata of fixed b_2 and $\dim \ker M_{d_j}$.



Freeness as a game

Build $\mathcal{A}$ **one line at a time** from an integer candidate pool $P \subset \mathbb{P}^2$.

- **state** — the partial arrangement (its lines + a global feature vector: b_2 , Δ , multiplicities, candidate exponents, current $\mathfrak{S}$);
- **action** — add the next line from P ;
- **reward** — combinatorial score (from Δ) + algebraic score ($1 - \mathfrak{S}$ when candidate exponents exist) + feasibility/shaping, and a terminal bonus for a verified-free arrangement.

A transformer policy/value network, trained with **PPO** + curriculum over (n, d_1, d_2) (to avoid collapse to the easy pair $(1, n - 2)$). Promising $\mathcal{A} \rightarrow$ **exact verification of Saito over $\mathbb{Q}$** .

The same skeleton as Part I — richer object, stronger solver.

Bootstrap for $n \geq 14$. RL works to $n = 13$; beyond, the action space is too large and the exact bonus unavailable. Use $\mathfrak{S}$ as a *pre-filter*: from a known free $\mathcal{A}$, keep candidate lines with $\mathfrak{S}(\mathcal{A} \cup \{L\}) < 0.05$, exact-verify survivors, and let level $n+1$ seed level $n+2$; remaining gaps by supersolvable constructions.

- $n \leq 13$: RL $\Rightarrow$ *hundreds* of verified free arrangements, every admissible exponent type (incl. balanced pairs).
- $14 \leq n \leq 20$: pre-filter + seeds $\Rightarrow$ every admissible (d_1, d_2) realized.

First application of ML to hyperplane-arrangement theory.

Open problems

Which problems are AI-suited? Open directions

Good candidates share three features: step-by-step construction, a *measurable proxy* for the target, and *exact verification* at the end.

- **Better relaxations.** Semicontinuous functionals like $\mathfrak{S}$ for other discrete invariants (near-freeness, the Bourbaki degree, ...).
- **Faster certifiers.** Exact symbolic verification is often the bottleneck.
- **Back to differential geometry.** G_2 / Calabi–Yau metrics (Lectures 2–3): where is the reward, and what plays the role of exact verification (the σ -measure, holonomy, ...)?
- **Scale.** Larger arrangements and higher $\mathbb{P}^r$; testing Terao's conjecture at scale.

Discovery proposes; verification disposes.

RL for line arrangements.

- _____, *A semicontinuous relaxation of Saito's criterion and freeness as angular minimization*, arXiv:2604.02995, 2026.
- Code & data: github.com/TomasSilva/FreeLineArrangements.

Free arrangements, Terao, Ziegler pairs.

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- H. Terao, *Generalized exponents of a free arrangement...*, Invent. Math., 1981.
- P. Orlik, H. Terao, *Arrangements of Hyperplanes*, Springer, 1992.
- M. DiPasquale, H. Schenck, et al.; DiPasquale–Sidman–Traves (grid $K_{3,n}$ scaffolding).
- A. Dimca, P. Pokora, works on nearly-free curves and Ziegler-type pairs.
- M. Barakat et al., *On the generation of rank-3 simple matroids...*, SIAM J. Discrete Math., 2021.

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- A. Davies et al., *Advancing mathematics by guiding human intuition with AI*, Nature, 2021.

- A. Fawzi et al., *Discovering faster matrix multiplication algorithms with RL*, *Nature*, 2022.
- S. Gukov, J. Halverson, F. Ruehle, P. Sułkowski, *Learning to unknot*, MLST, 2021.
- S. Gukov, J. Halverson, C. Manolescu, F. Ruehle, *Searching for ribbons with ML*, 2025.

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- J. Schulman et al., *Proximal policy optimization algorithms*, 2017.
- A. Vaswani et al., *Attention is all you need*, NeurIPS, 2017.

Thank You!

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